

Tensor Analysis

14. Tensor Form of Gradient, Divergence, and Curl

1. **Gradient:** If Φ is a scalar or invariant, the gradient of Φ is defined by

$$\nabla\Phi = \text{grad } \Phi = \Phi_{,p} = \frac{\partial\Phi}{\partial x^p}$$

where $\Phi_{,p}$ is the covariant derivative of Φ with respect to x^p .

2. **Divergence:** The divergence of A^p is the contraction of its covariant derivative with respect to x^{-q} , i.e. the contraction of $A^p_{,q}$. Then

$$\text{div } A^p = A^p_{,p} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k} (\sqrt{g} A^k)$$

3. **Curl:** The curl of A_p is $A_{p,q} - A_{q,p} = \frac{\partial A_p}{\partial x^q} - \frac{\partial A_q}{\partial x^p}$, a tensor of rank two. The curl is also defined as $-\epsilon^{pqr} A_{p,q}$.

4. **Laplacian:** The Laplacian of Φ is the divergence of grad Φ or

$$\nabla^2 \Phi = \text{div } \Phi_{,p} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^j} \left(\sqrt{g} g^{jk} \frac{\partial \Phi}{\partial x^k} \right)$$

In case $g < 0$, $\sqrt{g}$ must be replaced by $\sqrt{-g}$. Both cases $g > 0$ and $g < 0$ can be included by writing $\sqrt{|g|}$ in place of $\sqrt{g}$.